

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

POSPrinter (TM-U950/ TM-U375)

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Contents

SECTION 1. INTRODUCTION	1
SECTION 2. DETAILS ON SETTINGS	2
2.1 References of Firmware Versions	2
2.2 Settings of DIP Switches	2
2.3 Port Information	5
2.4 Device Settings	7
2.4.1 Usable Device Specific Settings.....	7
SECTION 3. FUNCTION DETAILS	8
3.1 Property Set Values and Default Values	8
3.1.1 Capability Set Values.....	8
3.1.2 List Properties.....	10
3.1.3 Width and Height Properties	11
3.1.4 Common Property Strings	12
3.1.5 PageMode Print Properties.....	12
3.2 Methods	13
3.3 Escape Sequences	14
3.4 Power Condition Reports.....	15
3.5 Synchronous Processing	15
3.6 Printing Positions	15
3.7 Electronic Logo Function (NVRAM)	15
3.8 Printable Bitmap Types and Sizes	16
3.9 Maintenance Counter	16
3.10 Automatic Recovery Function	16
SECTION 4. WARNINGS.....	17

Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-U950 and TM-U375 Series POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	Parallel	USB
TM-U950	TM-U950P	TM-U950U
TM-U950M ^{*1}	TM-U950PM ^{*1}	-
TM-U375	TM-U375P	-

^{*1} South Asia character model.

Before reading the manual, see the following explanation about the characteristic of the TM-U950 and TM-U375 models.

TM-U950 models

Station: Receipt / Journal / Slip (9-pin Serial impact dot matrix)

TM-U375 models

Station: Receipt / Slip (9-pin Serial impact dot matrix)

Rotate printing function

Throughout the manual, the various model names will be referred to as TM-U950 and TM-U375.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the TM-U950 and TM-U375 Series printers.

2.1 References of Firmware Versions

Refer to the release notes (Relnote.txt).

2.2 Settings of DIP Switches

Confirm that the following settings have been made correctly.

1) Serial port

* TM-U950

DIP-SW1

No.	Setting	
1	OFF	Fixed at OFF
2	OFF	Settable
3	OFF	Settable
4	OFF	Settable
5	OFF	Settable
6	OFF	Settable
7	OFF	Recommended
8	OFF	Fixed at OFF

DIP-SW2

No.	Setting	
1	OFF	Recommended
2	OFF	Recommended
3	ON	Fixed at ON
4	OFF	Recommended
5	OFF	Settable
6	ON	Fixed at ON
7	OFF	Fixed at OFF
8	OFF	Fixed at OFF

It is possible to change the settings of DIP-SW1-7 (Processing of the data input error) and DIP-SW2-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW1-8 (Handshake) to DTR/DSR.

Set DIP-SW1-1 (Bit length) to 8 bits.

Set DIP-SW1-2 to DIP-SW1-5 in accordance with the port information.

The described set values are the default values. For the details, refer to the product manual of the POSPrinter. Also, if these settings are changed, make sure to change the port information using the SetupPOS utility.

Set DIP-SW1-6 in accordance with whether or not a customer display is

connected. If connected, set to ON. If not, set to OFF.

It is possible to change the settings of DIP-SW2-1 (Auto line feed) and DIP-SW2-4 (Carriage Moving Speed), but it is recommended to leave them OFF.

Make other settings in accordance with the settings described above.

*** TM-U375**

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Recommended

Recommended

Fixed at OFF

Fixed at OFF

Settable

Settable

Settable

Settable

DIP-SW2

No.	Setting
1	OFF
2	OFF
3	OFF
4	??
5	??
6	??
7	OFF
8	OFF

Settable

Fixed at OFF

Settable

Not settable

Not settable

Not settable

Fixed at OFF

Fixed at OFF

It is possible to change the settings of DIP-SW1-1 (Processing of the data input error) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW1-3 (Handshake) to DTR/DSR.

Set DIP-SW1-4 (Bit length) to 8bits.

Set DIP-SW1-5 to DIP-SW1-8 in accordance with the port information.

The described set values are the default values. For the details, refer to the product manual of the POSPrinter. Also, if these settings are changed, make sure to change the port information using the SetupPOS utility.

Set DIP-SW2-1 in accordance with whether or not a customer display is connected. If connected, set to ON. If not, set to OFF.

DIP-SW2-4 to DIP-SW2-6 should not be change from current settings.

Make other settings in accordance with the settings described above.

2) Parallel Port / USB Port(TM-U950 only)*** TM-U950**

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF

DIP-SW2

No.	Setting
1	OFF
2	OFF
3	ON
4	OFF
5	OFF
6	ON
7	OFF
8	ON

Recommended
Recommended
Fixed at ON
Recommended
Settable
Fixed at ON
Fixed at OFF
Fixed at ON

It is possible to change the settings of DIP-SW2-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

It is possible to change the settings of DIP-SW2-1 (Auto Line Feed) and DIP-SW2-4(Carriage Moving Speed), but it is recommended to leave them OFF.

When parallel I/F is used with Windows 2000, Windows XP or Windows Vista please set Busy Condition of DIP-SW2-5 to ON (Buffer full).

Make other settings in accordance with the settings described above.

*** TM-U375**

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Recommended
Recommended
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF

DIP-SW2

No.	Setting
1	OFF
2	ON
3	OFF
4	??
5	??
6	??
7	OFF
8	ON

Fixed at OFF
Fixed at ON
Settable
Not settable
Not settable
Not settable
Fixed at OFF
Fixed at ON

It is possible to change the settings of DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

It is possible to change the settings of DIP-SW2-1 (Auto line feed) and DIP-SW2-4 (Carriage Moving Speed), but it is recommended to leave them OFF.

Make other settings in accordance with the settings described above.

2.3 Port Information

1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	1200,2400,4800,9600
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	2 to 64 (TM-U375) 32 to 1024 (TM-U950)
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	9600
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	10 (TM-U375) 1024 (TM-U950)
Output interval time [ms]	2500

2) Port information when using parallel port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	2 to 64 (TM-U375) 32 to 1024 (TM-U950)
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	10 (TM-U375) 1024 (TM-U950)
Output interval time [ms]	2500

2.4 Device Settings

The following explanation is about the settings for TM-U950 and TM-U375.

2.4.1 Usable Device Specific Settings

For the TM-U950 and TM-U375, the following Device Specific Settings are settable by the SetupPOS utility. For the detail, please refer to the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Ignore firmware version check
	Homogenize Error Codes
	Output complete timeout
Color Bitmap	Method
	Brightness
	Primary
Status Log	ERROR
	OFFLINE
	Log file name (full path name)
	Maximum file size [KB]

Section 3. Function Details

This section describes the functions of the TM-U950 and TM-U375 printers in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

3.1 Property Set Values and Default Values

The following explanation is about the property set values and the default values.

3.1.1 Capability Set Values

The following values are the Capability set values.

Capability Name	TM-U950	TM-U375
CapTransaction	TRUE	TRUE
CapCoverSensor	TRUE	TRUE
CapConcurrentRecSlp	FALSE	FALSE
CapConcurrentJrnSlp	FALSE	FALSE
CapConcurrentJrnRec	TRUE	FALSE
CapConcurrentPageMode	FALSE	FALSE
CapCharacterSet	PTR_CCS_ASCII	PTR_CCS_ASCII
CapMapCharacterSet	FALSE	FALSE
CapJrnUnderline	TRUE	FALSE
CapJrnNearEndSensor	TRUE	FALSE
CapJrnItalic	FALSE	FALSE
CapJrnEmptySensor	TRUE	FALSE
CapJrnDwideDhigh	TRUE	FALSE
CapJrnDwide	TRUE	FALSE
CapJrnDhigh	TRUE	FALSE
CapJrnColor	PTR_COLOR_PRIMARY	0
CapJrnCartridgeSensor	0	0
CapJrnBold	TRUE	FALSE
CapJrn2Color	FALSE	FALSE
CapJrnPresent	TRUE	FALSE
CapRecPageMode	FALSE	TRUE
CapRecUnderline	TRUE	TRUE
CapRecStamp	FALSE	FALSE

CapRecRotate180	TRUE	TRUE
CapRecRight90	FALSE	TRUE
CapRecPapercut	TRUE	FALSE
CapRecNearEndSensor	TRUE	TRUE
CapRecMarkFeed	0	0
CapRecLeft90	FALSE	TRUE
CapRecItalic	FALSE	FALSE
CapRecEmptySensor	TRUE	TRUE
CapRecDwideDhigh	TRUE	TRUE
CapRecDwide	TRUE	TRUE
CapRecDhigh	TRUE	TRUE
CapRecColor	PTR_COLOR_PRIMARY	PTR_COLOR_PRIMARY
CapRecCartridgeSensor	0	0
CapRecBold	TRUE	TRUE
CapRecBitmap	TRUE	TRUE
CapRecBarCode	FALSE	FALSE
CapRec2Color	FALSE	FALSE
CapRecPresent	TRUE	TRUE
CapSlpUnderline	TRUE	TRUE
CapSlpRotate180	TRUE	TRUE
CapSlpRight90	FALSE	TRUE
CapSlpNearEndSensor	TRUE	TRUE
CapSlpLeft90	FALSE	TRUE
CapSlpItalic	FALSE	FALSE
CapSlpEmptySensor	TRUE	TRUE
CapSlpDwideDhigh	TRUE	TRUE
CapSlpDwide	TRUE	TRUE
CapSlpDhigh	TRUE	TRUE
CapSlpColor	PTR_COLOR_PRIMARY	PTR_COLOR_PRIMARY
CapSlpCartridgeSensor	0	0
CapSlpBothSidesPrint	FALSE	FALSE
CapSlpBold	TRUE	TRUE
CapSlpBitmap	TRUE	TRUE
CapSlpBarCode	FALSE	FALSE
CapSlp2Color	FALSE	FALSE
CapSlpFullslip	TRUE	TRUE
CapSlpPresent	TRUE	TRUE
CapSlpPageMode	FALSE	TRUE

3.1.2 List Properties

The List Properties are explained in the following.

*TM-U950

List Property	Settings
CharacterSetList	(TM-U950) "255,437,850,858,860,863,865,998"
	(TM-U950M) ^{*1} "120,121,123,126,437,998" ^{*2}
JrnLineCharsList	"30,40"
RecLineCharsList	"30,40"
SlpLineCharsList	"66,88"
RecBarCodeRotationList	""
RecBitmapRotationList	"0, 180"
SlpBarCodeRotationList	""
SlpBitmapRotationList	"0, 180"
FontTypefaceList	""

^{*1} Device is limited to South Asia model.

^{*2} The list is for South Asia character model.

*TM-U375

List Property	Settings
CharacterSetList	"437,850,858,860,863,865,998"
JrnLineCharsList	""
RecLineCharsList	"33,40"
SlpLineCharsList	"33,40"
RecBarCodeRotationList	""
RecBitmapRotationList	"0,R90, L90, 180"
SlpBarCodeRotationList	""
SlpBitmapRotationList	"0,R90, L90, 180"
FontTypefaceList	""

3.1.3 Width and Height Properties

The width and height properties are described below.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	12	127	0
JrnLineSpacing ^{*1}	12	127	0
SlpLineSpacing	12	127	0
SlpLineHeight [dot]	9		
RecLineHeight [dot]	9		
JrnLineHeight [dot] ^{*1}	9		
SlpLineWidth [dot]	400(TM-U950), 200(TM-U375)		
RecLineWidth [dot]	180(TM-U950), 200(TM-U375)		
JrnLineWidth [dot] ^{*1}	180		
RecSidewaysMaxLines	0(TM-U950), 16(TM-U375) ^{*3}		
RecSidewaysMaxChars ^{*1}	0(TM-U950), 117(TM-U375) ^{*4}		
RecLinesToPaperCut	11(TM-U950), 9(TM-U375) ^{*2}		
SlpSidewaysMaxLines	0(TM-U950), 16(TM-U375) ^{*3}		
SlpSidewaysMaxChars	0(TM-U950), 117(TM-U375)		
SlpMaxLines	0		

X : No settings

^{*1} Available only for the TM-U950. On the other models, there is no setting.

^{*2} It can be changed by the settings of the RecLineSpacing or the character height.

When the RecLineSpacing is "0", the RecLinesToPaperCut is "-1".

^{*3} It can be changed by the settings of the XxxLineSpacing or the XxxLineHeight.

^{*4} It can be changed by the settings of the font width.

*When the RecLineSpacing is 0, the RecLinesToPaperCut is -1.

*When the XxxLineSpacing is 0, the XxxSidewaysMaxLines is -1.

3.1.4 Common Property Strings

The Device information properties are described below.

- TM-U950

I/F	DeviceName	DeviceDescription
S	TM-U950	EPSON TM-U950 POS Printer
	TM-U950M	EPSON TM-U950M POS Printer
P	TM-U950P	EPSON TM-U950P POS Printer
	TM-U950PM	EPSON TM-U950PM POS Printer
U	TM-U950U	EPSON TM-U950U POS Printer

- TM-U375

I/F	DeviceName	DeviceDescription
S	TM-U375	EPSON TM-U375 POS Printer
P	TM-U375P	EPSON TM-U375P POS Printer

I/F indicate the connected interface.

The following is the list of the two connecting interfaces.

S: Serial

P: Parallel

U: USB

3.1.5 PageMode Print Properties

The Device information properties are described below.

- TM-U375

Property	Station ^{*2}		
	Receipt	Slip	Slip (Validation)
PageModeArea	"200", "704"	"200", "704"	"200", "704"
PageModeDescriptor ^{*1}	BM/BMR	BM/BMR	BM/BMR

^{*1} Following setting values are used for the PageModeDescriptor property.

BM : Bitmap printing is available.

BC : Barcode printing is available.

BMR : Rotated printing of bitmap is available.

BCR : Rotated printing of barcode is available.

^{*2} If the Station's CapRecPageMode and/or CapSlpPageMode property values are FALSE, the PageModeArea property shall have "" and the PageModeDescriptor property shall have "0" respectively as a setting value.

3.2 Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal ^{*1}	O	X
PrintImmediate	O	O ^{*2}
PrintBarCode	X	O ^{*3}
PrintBitmap	O	O ^{*4}
PrintMemoryBitmap	O	O ^{*4}
CutPaper ^{*1}	O (1-79: Three points remain uncut.) (80-100: One point remains uncut.)	X
MarkFeed	X	X
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	O	O
BeginRemoval	O	O
EndInsertion	O	O
BeginInsertion	O	O
ClearPrintArea	O	O
PageModePrint	O	O

O : Supported

X : Unsupported

^{*1} Available only for the TM-U950. On the other models, there is no setting.

^{*2} If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

^{*3} If other than "LEFT" is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

^{*4} If other than "LEFT" is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

3.3 Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

Escape Sequence	TM-U950	TM-U375	Compatibility with the PageMode printing
#P	0~100	X	X
#fP	0~100	X	X
#sP	X	X	X
sL	X	X	X
#B	O	O	O
tL	O	O	O
bL	O	O	O
#R	X	X	X
#fF	0-9999	0~9999	O
#uF Base Pitch [inch]	0-approx. 50 cm	0~ approx. 50 cm	O
#rF Maximum [inch]	1	X	X
#E	0~65535	0~65535	X
#fT	X	X	X
bC	O	O	O
!bC	O	O	O
#uC	1	1	O
iC	X	X	X
!iC	X	X	X
#rC	1	1	O
rvC	X	X	X
!rvC	X	X	X
#sC	X	X	X
#fC	X	X	X
tbC	X	X	X
!tbC	X	X	X
tpC	X	X	X
!tpC	X	X	X
1C	O	O	O
2C	O	O	O
3C	O	O	O
4C	O	O	O
#hC	1-2	1-2	O
#vC	1-2	1-2	O
cA	O	O	O ^{*1}
rA	O	O	O ^{*1}
lA	O	O	O

N	O	O	O
---	---	---	---

O : Supported

X : Unsupported

Numbers: Settable range

^{*1} Regardless of the PageModeHorizontalPosition property setting, center or right adjust what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

3.4 Power Condition Reports

The TM-U950 and TM-U375 models do not support Power Condition Reports.

3.5 Synchronous Processing

The TM-U950 and TM-U375 models do not use Process ID to determine output completion.

3.6 Printing Positions

The TM-U950 and TM-U375 models support the function for setting printing position.

Function	TM-U950	TM-U375
Left margin	X	O
Printing Position	O	O

O: Supported

X: Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.7 Electronic Logo Function (NVRAM)

The TM-U950 and TM-U375 models do not support the Electronic Logo Function.

3.8 Printable Bitmap Types and Sizes

The TM-U950 and TM-U375 models support the following bitmap commands. For the detail, please refer to the Section 3 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”. The allowance ranges for bitmaps are as follows.

- TM-U950

Bitmap command type	Allowance range		
Download bitmap	x (dot)	y (dot)	xy
	1-2040	1-384	<= 98304
One-line bitmap	No setting range		

- TM-U375

Bitmap command type	Allowance range		
Download bitmap	x (dot)	y (dot)	xy
	1-2040	1-2040	<= 32768
One-line bitmap	No setting range		

3.9 Maintenance Counter

The TM-U950 and TM-U375 models do not support the Maintenance Counter.

3.10 Automatic Recovery Function

The TM-U950 and TM-U375 models do not have a function for automatic recovery when the power is turned on again after an interruption of power.

Section 4. Warnings

This section describes precautions in use of TM-U950 and TM-U375.

- Use the PTR_TWO_RECEIPT_JOURNAL when the print data is output both on the Receipt and on the Journal, in order to keep a high compatibility in the PrintTwoNormal method. The PTR_S_JOURNAL_RECEIPT is still available to be compatible with the old version. The operation with the PTR_S_JOURNAL_RECEIPT is same as the PTR_TWO_RECEIPT_JOURNAL.